When Graph-Derived Neighbourhood Features Do Not Help: A Leakage-Controlled Evaluation of Behavioural-LRFM Churn Prediction in Non-Contractual Retail

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Abstract

Customer similarity graphs are an appealing addition to RFM-style churn models. If churn spreads through behavioural peer groups, the churn rate among a customer's nearest neighbours should help predict that customer's own churn. We test that idea on the Online Retail II dataset under two controls. First, the neighbourhood and community churn rates are recomputed inside every training fold, so no test label reaches the feature matrix. Second, every comparison runs over ten repetitions of five-fold cross-validation and is judged by the corrected resampled t-test rather than by a single split. Under these controls the idea does not hold. No feature tier beats a plain three-variable RFM baseline, on either ROC-AUC or PR-AUC, for either classifier. The finding holds as the graph degree varies from 5 to 20. The graph-enhanced tier gains +0.0033 ROC-AUC on one fold partition and reverses sign across ten. The only significant comparison points the other way: logistic regression on the three raw RFM variables beats the 38-feature gradient-boosted graph model ($p = 0.010$). Attribution and contribution also come apart, since the neighbourhood churn rate ranks third by SHAP yet removing it costs nothing. We report the pipeline, the controls and the negative outcome, and argue that single-split evaluation cannot separate an effect this small from fold noise.

Keywords: Churn prediction · RFM · Customer similarity graph · Community detection · Repeated cross-validation · Negative result.

1 Introduction

Non-contractual retail gives no cancellation signal. A customer who stops buying does not announce it, so churn has to be defined by inactivity over a chosen horizon, and the modelling task is to predict that inactivity from purchase history. The RFM representation—recency, frequency, monetary value—has been the working vocabulary of this problem for decades, and a large applied literature extends it: adding relationship length, adding demographic attributes [9], or unrolling the triple over time [14].

A more recent line treats the customer base as a graph rather than a table. Corrêa Vianna Filho et al. formulate RFM segmentation as a maximum

k-cut problem on a customer graph, where identical score vectors collapse into super-vertices and the reduced graph is bounded independently of the number of customers [6]. Once customers are vertices, a next step suggests itself: if a similarity graph can be built, then community detection [1, 17] can partition it, and graph-derived quantities—degree, centrality, and in particular the churn rate observed among a customer's neighbours—can be appended to the feature matrix of a supervised churn model. The underlying hypothesis is a contagion one: churn risk is partly a property of the behavioural neighbourhood, not only of the individual.

This paper tests that hypothesis and does not find support for it. We build the pipeline the hypothesis calls for—a nine-dimensional behavioural extension of RFM, a k-nearest-neighbour customer similarity graph, Louvain and Leiden communities, and graph node features including neighbour and community churn rates—and we evaluate it against progressively simpler feature sets. The evaluation is where the contribution lies. Two controls turn out to be decisive:

1. **Fold-local recomputation of the neighbourhood features.** Both are aggregates of the target, so computing them once over the full dataset places test labels inside the training features. We recompute them inside every training fold from training labels only.
2. **Repeated cross-validation with a variance-corrected test.** A single split gives a point estimate with no uncertainty attached. We use ten repetitions on a shared fold partition and report the corrected resampled t-test of Nadeau and Bengio [15], which adjusts for the training-set overlap that makes the naive paired test anti-conservative [8, 2].

With both controls in place, the effect the pipeline was built to demonstrate disappears. We state our contributions conservatively:

- A negative result, reported as such: neither the behavioural extension nor the graph features improve discrimination over a three-variable RFM baseline, at any graph degree tested, and the only significant comparison favours the simplest model in the study.
- An evaluation protocol for this application area, together with the observation that differences below roughly 0.005 ROC-AUC on this dataset are distinguishable neither from fold noise nor from library-version drift.
- A dissociation between SHAP attribution rank and predictive contribution, which we state explicitly because feature-importance rankings are frequently read as evidence of usefulness.

2 Related Work

Three strands meet in this problem, and the tension between them is what makes a negative result informative rather than merely disappointing.

The *representation* strand asks what to measure about a customer. RFM persists because it is cheap and interpretable, and the extensions that follow

it generally add dimensions rather than replace the idea: relationship length alongside the original three, demographic attributes [9], or a temporal unrolling in which the RFM triple becomes a sequence rather than a point [14]. What these extensions share is an implicit claim that the added dimensions carry information the original three do not. That claim is testable, and it is one of the things we test.

The *structural* strand asks whether customers should be modelled jointly rather than independently. Vianna Filho et al. make the strongest form of this argument for segmentation: by recasting RFM segmentation as maximum k-cut over a customer graph, the segmentation objective becomes an explicit graph quantity rather than a by-product of a distance-based clustering heuristic [6]. Their work is the starting point of ours, but it stops at segmentation and does not address supervised prediction. Community detection supplies the machinery for a supervised extension: modularity-based partitioning [16, 1] and its refinement guaranteeing well-connected communities [17] are standard, well-implemented, and apply directly to a similarity graph over customers. The step from there to appending the neighbourhood churn rate as a feature is short, which is precisely why it deserves scrutiny: that feature is an aggregate of the label, and features of this kind are the classic vehicle for target leakage.

The neighbourhood-churn feature is not itself new, and the settings where it has worked are informative. Telecommunications studies build it from call-detail records, where an edge means two subscribers actually spoke, and report that a subscriber's risk rises with the number of churned neighbours [11]; more recent work treats the social graph as a temporal object and predicts churn from how it evolves [12]. What separates those settings from ours is what an edge means. Theirs records an interaction that exists independently of the model's feature matrix; ours records proximity within the very feature space the classifier already sees. That difference turns out to matter, and Sect. 9.2 returns to it.

The *evaluation* strand is where the applied literature is weakest and where the methodological literature has long been settled. Overlapping resampled training sets inflate Type I error in the tests normally used to compare learning algorithms [8]; the variance correction that repairs the paired test for exactly this case has been available for two decades [15], along with evidence on how such tests replicate in practice [2, 7]. Applied churn studies, including our own earlier reporting of the present pipeline, nonetheless rank configurations by a single cross-validated point estimate. At differences on the order of 10^{-3} ROC-AUC that cannot separate an effect from a fold partition. Our results are an instance of the gap between these two strands, not a criticism of any individual study.

Gradient-boosted trees [5, 10] and additive feature attribution [13] are used here as standard tools, not as objects of study.

3 Pipeline Overview

Figure 1 shows the stages and where the two controls sit. The left-to-right upper row builds the representation; the lower row, read right to left, carries it through

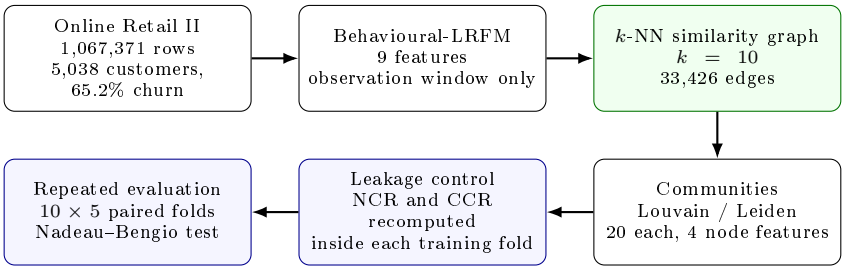


Fig. 1. Analysis pipeline. Shaded stages are the two controls that determine the result: label-derived neighbourhood features are recomputed inside every training fold, and every comparison is run over ten repetitions of five-fold cross-validation with a variance-corrected significance test.

the controls into the evaluation. The two shaded stages are the ones that decide the outcome: the leakage control that keeps label-derived features inside the training fold, and the repeated evaluation that attaches uncertainty to every comparison.

4 Data and Feature Construction

4.1 Dataset and Windows

We use Online Retail II, the transaction log of a UK-based non-store retailer covering December 2009 to December 2011, distributed through the UCI Machine Learning Repository [3] and introduced in the RFM segmentation study of Chen et al. [4]. The raw file contains 1,067,371 transaction rows across two worksheets (525,461 and 541,910); the 541,000 figure often quoted for this dataset is the second worksheet alone.

Features and labels are separated in time. The observation window runs from 2009-12-01 to 2011-06-30 and is the only period from which features are computed. The prediction window runs from 2011-07-01 to 2011-09-30; a customer active in the observation window is labelled a churner if no purchase appears in the prediction window. After filtering to valid transactions inside the observation window (556,212 rows) and aggregating to the customer level, the working matrix contains 5,038 customers. The churn rate is 65.2%.

That last number deserves emphasis, because it inverts an assumption often carried into this problem. Churners are the *majority* class here, not a rare positive to be rescued by resampling. A 90-day inactivity horizon applied to a retailer with long interpurchase intervals labels most of the base as churned. This makes PR-AUC less informative than it would be under genuine imbalance, and we report it alongside ROC-AUC rather than as the primary metric.

4.2 Behavioural-LRFM Features

Nine customer-level features are computed inside the observation window: length L (days between first and last purchase), recency R, frequency F (distinct invoices), monetary value M, average order value, basket size, product diversity, cancellation rate, and mean interpurchase time. Cancellation rate is restricted to cancellations dated inside the observation window; an earlier version of this pipeline computed it over the full log, which allowed post-window cancellations to enter a pre-window feature.

5 Graph Construction and Community Detection

The nine features are standardised and a k-nearest-neighbour graph is built over customers with $k = 10$ and Euclidean distance, with edge weight $w(i, j) = 1/(1+d(i, j))$. The resulting graph has 5,038 nodes and 33,426 edges. Louvain [1] yields 20 communities with modularity 0.8385; Leiden [17] yields 20 communities with modularity 0.8323.

The high modularity should not be over-read: a k-NN graph with fixed k imposes near-uniform degree and strong local clustering by construction, so a value above 0.8 reflects how the graph was built at least as much as any structure in the customer base.

Six graph-derived features are appended: degree, weighted degree, PageRank, clustering coefficient, and the two neighbourhood aggregates that motivate the design—neighbour churn rate (mean label among a customer's graph neighbours) and community churn rate (mean label within the customer's Louvain community).

6 Leakage Controls

Both neighbourhood aggregates are functions of the target. Computing either over the full dataset before cross-validation lets the label of a test customer influence the training features of its neighbours, and the resulting optimism can be substantial. We therefore recompute both inside each training fold: for customer i, the neighbour churn rate averages labels over $\mathcal{N}(i) \cap \mathcal{T}$, where $\mathcal{T}$ is the training index set, falling back to the training-fold base rate when that intersection is empty; the community churn rate is computed the same way within each Louvain community. The graph and the communities themselves are built from features only, never from labels, and are therefore fixed across folds.

Three further controls apply throughout: standardisation is fitted inside the cross-validation pipeline rather than on the full matrix; K-means and Louvain cluster identifiers are one-hot encoded rather than passed as integers; and both Louvain and Leiden are run, so that the choice of community detector is not a hidden degree of freedom.

7 Evaluation Protocol

7.1 Feature Tiers

Five nested tiers isolate the contribution of each modelling layer: **RFM** (R, F, M); **LRFM** (adding L); **B-LRFM** (the nine behavioural features); **B-LRFM+Seg** (adding 23 one-hot segment and community indicators); and **Graph-Enhanced**, the full 38-feature matrix adding the four static graph features and the two fold-local neighbourhood aggregates. Two classifiers are evaluated on every tier: LightGBM [10] as the primary model and ℓ_2-regularised logistic regression as an interpretable reference.

7.2 Repeated Cross-Validation and Significance

We run ten repetitions of stratified five-fold cross-validation, giving 50 folds. Within a repetition all five tiers and both classifiers see the *same* fold partition, so per-fold differences are properly paired. For a difference $d_1, \ldots, d_n$ over $n = 50$ folds we report the corrected resampled t-statistic [15]

$$t = \frac{\bar{d}}{\sqrt{\hat{\sigma}^2 \left(\frac{1}{n} + \frac{n_2}{n_1} \right)}}, \qquad \frac{n_2}{n_1} = \frac{1}{4}, \tag{1}$$

where n_2/n_1 is the ratio of test-set to training-set size, which for five-fold cross-validation is $1/4$. We report p_{NB} from this statistic only. Bootstrap confidence intervals over the 50 differences are given for effect magnitude, but we caution against reading them as significance statements: the bootstrap resamples the differences as if independent, while repeated folds share training data, so those intervals are narrower than the correlation structure warrants. This is the same overlap that makes the naive paired t-test anti-conservative [8].

Run with the single seed used by the original pipeline, the significance harness reproduces that pipeline's per-tier ROC-AUC to four decimal places on all five tiers, so any divergence below is attributable to the fold partition rather than to a re-implementation difference.

8 Results

8.1 One Split Says the Opposite of Ten

Table 1 puts the two evaluations side by side, and the contrast is the result of this paper. On the single stratified five-fold split this pipeline originally reported, Graph-Enhanced is the best tier, +0.0033 ROC-AUC above RFM. Over ten repetitions it is jointly the worst, −0.0021 below RFM. Nothing about the model changed between those two columns; only the fold partition did. Even at the seed that produces it, the +0.0033 gap has a bootstrap 95% interval of $[-0.0025, +0.0099]$, which contains zero.

Table 1. LightGBM across the five feature tiers. The single-split column is the seed-42 evaluation this pipeline first reported; the remaining columns are ten repetitions of the same protocol. p_{NB} is the corrected resampled t-test against RFM (Sect. 7.2); no tier reaches significance. Why the single-split column misleads is taken up in Sect. 9.

	Single split	Repeated CV (10×5 folds, paired)		
Feature tier	ROC-AUC	ROC-AUC	Δ vs RFM	p_{NB}
RFM (baseline)	0.8065	0.8066	—	—
LRFM	0.8071	0.8074	+0.0009	0.755
B-LRFM	0.8067	0.8054	−0.0012	0.742
B-LRFM+Seg	0.8073	0.8044	−0.0022	0.561
Graph-Enhanced	**0.8098**	0.8044	−0.0021	0.590

Table 2. Cross-model comparisons over the same 50 paired folds. The intervals are bootstrap and are given for magnitude only; significance is read from p_{NB}, for the reason set out in Sect. 7.2.

Comparison	Δ ROC-AUC	95% CI	p_{NB}
LogReg / B-LRFM vs. LGBM / Graph-Enh.	+0.0113	[+0.0096, +0.0130]	**0.001**
LogReg / RFM vs. LGBM / Graph-Enh.	+0.0099	[+0.0080, +0.0119]	**0.010**
LogReg / RFM vs. LGBM / RFM	+0.0078	[+0.0056, +0.0100]	0.068

No tier differs significantly from the RFM baseline, and three of the four point differences are negative. The same holds for PR-AUC and for logistic regression as the classifier: across the four metric-classifier combinations—sixteen tier-versus-baseline comparisons—the smallest p_{NB} is 0.144. On the single split the other metrics agree with ROC-AUC, with Graph-Enhanced leading on PR-AUC (0.8689), F1 (0.8108) and recall (0.8009); that agreement is what makes a single split persuasive and is exactly why it is not sufficient.

8.2 The One Significant Effect

Table 2 reports the comparisons that do reach significance. All of them favour logistic regression over the gradient-boosted graph model.

A three-variable linear model outperforms a 38-feature gradient-boosted model built on a customer similarity graph. The margin is roughly three times the +0.0033 gap that the single-split evaluation reported as the pipeline's improvement, and larger in magnitude than any difference between feature tiers. We draw a narrow conclusion from this: on this dataset, at this sample size, with this churn definition, the discriminative signal is close to exhausted by recency, frequency and monetary value, and additional model capacity is spent on variance rather than on signal.

Figure 2 shows the whole result in one picture. Each line is flat: moving along it—adding tenure, then behaviour, then segments, then the graph—changes nothing visible. The vertical distance between the two lines, which is only a

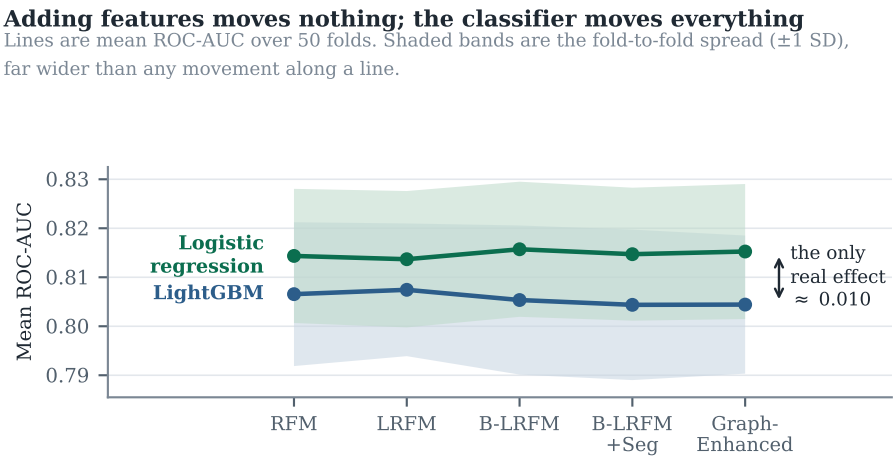


Fig. 2. Mean ROC-AUC across the five feature tiers for both classifiers. Bands show ±1 standard deviation of the 50 fold scores, that is, the spread an individual split is drawn from. Uncertainty on the paired differences themselves is in Tables 1 and 2.

Table 3. Graph-Enhanced versus the RFM baseline as the graph degree varies. LightGBM, 10×5 paired folds at every setting; the graph, the Louvain communities and the fold-local features are rebuilt for each k.

k	Edges	Communities	Δ vs RFM	p_{NB}
5	16,902	31	−0.0006	0.880
10	33,426	20	−0.0021	0.590
15	50,054	16	+0.0017	0.666
20	66,652	16	−0.0027	0.500

change of classifier, is the one real effect. The shaded bands give the reason a single split cannot settle this: fold-to-fold spread is about 0.014 standard deviation, an order of magnitude larger than any difference between tiers.

8.3 The Result Does Not Depend on the Graph Degree

A negative result about a graph invites the obvious question of whether it is a result about one particular graph. We therefore rebuilt the graph, the communities and the fold-local features from scratch at $k \in \{5, 10, 15, 20\}$ and reran the full repeated-CV protocol at each setting. Table 3 reports the outcome.

Quadrupling the edge count and halving the number of communities changes the graph substantially and the conclusion not at all: no setting reaches significance, and the sign of the difference is not even stable, turning positive at $k = 15$ and negative at the three others. A quantity whose sign depends on a

tuning parameter it should not depend on is behaving like noise, which is what the p-values say directly.

8.4 Attribution Is Not Contribution

Global SHAP values [13] on the graph-enhanced model rank recency first (1.087) and monetary value second (0.560), followed by neighbour churn rate (0.250), interpurchase time (0.224) and community churn rate (0.215). Read as an importance ranking, this appears to confirm that the neighbourhood aggregates matter: two of the top five features are graph-derived.

They do not matter for discrimination: the tier without them (B-LRFM+Seg, 0.8044) and the tier with them (Graph-Enhanced, 0.8044) are indistinguishable. This is a matter of definition rather than a contradiction. SHAP attributes how a *fitted* model distributes credit across the features it was given, so a feature correlated with both the target and the rest of the feature set absorbs attribution while adding nothing the others did not already carry. Attribution rank answers "what is this model using?"; only an ablation answers "would removing it cost anything?", and here it answers no.

8.5 Relation to the Segmentation Baseline

Our starting point [6] segments customers on discrete 1–5 RFM scores in three dimensions; the present work predicts churn in a nine-dimensional continuous space. Silhouette scores are not comparable across different feature spaces—a low-dimensional discrete space produces tighter clusters as a property of the space—so we make no head-to-head clustering-quality claim. For reference, on its own scoring space the baseline reaches a silhouette of 0.398 at $k = 4$ as published, which we reproduce at 0.398; K-means on our nine-dimensional space reaches 0.219 and Louvain reaches -0.050 on the same measure. The relationship between the two works is scope extension, not improvement. The communities remain usable for prioritisation even though they do not improve prediction: ranked by churn probability times monetary value, they separate a high-value at-risk group (327 customers, 82% churn) from a high-value loyal one (418 customers, 24% churn). With no campaign-response labels in the data, that is a descriptive output only.

9 Discussion

9.1 Effect Sizes Below the Noise Floor

Two independent sources of variation are the same size as the effects the pipeline was built to detect. The fold partition contributes roughly 0.003 ROC-AUC, as the sign reversal above shows. Library versions contribute as much: re-running the auxiliary improvement pipeline under newer libraries moved its best configuration by -0.0033 ROC-AUC and the Leiden community count from 20 to 21.

We therefore treat differences below roughly 0.005 ROC-AUC on this dataset as uninformative, and suggest that comparisons here either report a variance-corrected test or refrain from ranking configurations inside that band.

9.2 Why the Contagion Hypothesis Might Still Hold

Our result is a failure to reject a null on one dataset with one graph construction, not a demonstration that behavioural peer effects are absent. The most likely explanation is that we built the wrong graph: its edges encode proximity in a feature space derived from the same purchase history that already feeds the classifier, so the neighbour churn rate is plausibly a smoothed restatement of what the model has rather than new information. This reading is consistent with where the feature does work. The telecommunications and social graph results cited in Sect. 2 are built on recorded interaction—who called whom, who plays with whom—not on similarity computed from the model's own inputs [11, 12]. A retail graph built from an interaction signal absent from the feature matrix—shared baskets, shared categories, referral or household links—would be a genuinely different test, and is the experiment we would run next. Two weaker readings also fit: a 90-day inactivity label on a majority-churn base may be too coarse for neighbourhood effects to surface, and with all tiers near 0.81 ROC-AUC the binding constraint may be the representation rather than the model class.

10 Threats to Validity

Single dataset. All results come from one retailer over two years. We make no claim about other sectors or horizons; the effect-size floor we report is specific to this dataset and sample size.

One graph family. The degree sweep in Sect. 8 shows the result is not an artefact of k, but every setting is still a k-nearest neighbour graph on standardised Behavioural-LRFM features with the same distance and the same weighting. A different graph family—built from co-purchase, category overlap or household links—is untested here and is where we expect the hypothesis to get a fairer hearing.

Planned components not implemented. Two components described in our study design were not implemented and form no part of any result reported here: a country-of-origin categorical encoding, and spectral clustering as a third community detection algorithm. We list them so that the gap between design and execution is visible.

Metric choice. With churners in the majority, PR-AUC does not carry its usual interpretation. We report both metrics and base no conclusion on PR-AUC alone.

Literature search is targeted, not exhaustive. We searched for prior use of neighbourhood-churn features and located the telecommunications and social graph work discussed in Sect. 2, which uses interaction graphs rather than

similarity graphs. We did not find the retail similarity-graph variant tested here, but the search was targeted rather than systematic, and we make no priority claim of any kind.

No business validation. Expected retention value ranks customers by a model quantity, not by observed campaign response. Its practical value is untested.

11 Conclusion

We built the pipeline the behavioural-peer-effect hypothesis calls for and evaluated it under fold-local recomputation of the label-derived features and repeated cross-validation with a corrected test. The hypothesis is not supported on this dataset at any graph degree tested, and the only significant comparison favours logistic regression on three raw RFM variables over the full graph model ($p = 0.010$).

The methodological point generalises further than the negative result. An evaluation that reports one cross-validated number per configuration cannot distinguish a 0.003 ROC-AUC difference from the partition that produced it, and differences of that size are what feature-engineering studies in this area frequently report. The controls used here are inexpensive—ten repetitions of five-fold cross-validation on 5,038 customers is a matter of minutes—and we would argue they belong in the default protocol rather than in an appendix.

Reproducibility. Code, per-fold scores, significance summaries and raw run logs are available at `https://github.com/VoDucNhatku/graph-behavioral-lrfm-churn`, tagged `v1.0-submission`. Library versions are pinned to the environment that produced every reported number, and the result figure is generated from the committed per-fold scores, so it cannot drift from the tables. The full pipeline runs in under ten minutes on a laptop with no GPU.

Declaration on generative AI. During the preparation of this work the author used a large language model (Claude, Anthropic) to draft and revise the manuscript, to implement parts of the evaluation and figure code, and to cross-check reported numbers against the result files. The author verified every reported number against the archived runs and takes full responsibility for the content.

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